AI Automation Engineering

with Generative AI, Agentic AI, LangGraph & MCP

Redesigned 8 Half-Day Enterprise Training Program for Insurance Domain UiPath Developers

Duration	**8 Half Days**
Total Hours	32 Hours
Daily Schedule	4 Hours per Day
Audience	UiPath Developers, RPA Developers, Automation Leads, Solution Architects
Domain Focus	Insurance Claims, Policy Servicing, Underwriting, Customer Support, Fraud Review
Delivery Mode	Instructor-led, lab-driven, project-based training

Focus: 80% hands-on labs and implementation, 20% concept explanation

1. Program Objective

The objective of this redesigned program is to transform UiPath developers into AI Automation Engineers who can build insurance-domain automation solutions using Python, Generative AI, LangChain, LangGraph, MCP, APIs, and UiPath integration. **The program is intentionally practical and implementation-focused.**

Reference Basis: Redesigned from the attached 5-day / 40-hour AI Automation Engineering outline. The revised version reduces duration to 8 half days / 32 hours and increases focus on Python, Generative AI, Agentic AI, LangGraph, MCP, UiPath integration, and insurance-domain hands-on implementation.

2. Target Audience

- UiPath Developers and RPA Developers
- Automation Anywhere / Blue Prism developers moving into AI Automation
- Insurance automation teams
- Solution Architects and Automation Leads
- Digital Transformation and Intelligent Automation teams

3. Recommended Learning Outcomes

- Use Python for insurance automation, document processing, data validation, APIs, and JSON workflows.
- Apply Generative AI for policy summarization, claim summarization, email generation, classification, and decision support.
- Build LangChain-based assistants for policy Q&A, customer support, and claims processing.
- Design Agentic AI workflows using LangGraph with state, nodes, conditional routing, human review, and error handling.
- Understand and implement MCP concepts for tool discovery and AI-driven enterprise integration.
- Integrate Python, LLMs, APIs, and UiPath robots into insurance automation workflows.
- Build an end-to-end insurance claims automation capstone using AI agents and UiPath orchestration.

4. Module-Wise Duration Distribution

Module	Duration	Primary Focus
Python for AI Automation	6 Hours	Insurance data handling, APIs, JSON, Excel, exception handling
Generative AI & Prompt Engineering	4 Hours	Structured prompts, claim summaries, policy summaries, email generation
AI APIs & Insurance Automation	2 Hours	OpenAI / Azure OpenAI SDK, secure API usage, response parsing
LangChain Fundamentals	3 Hours	Prompt templates, chains, parsers, tools, assistants
Agentic AI using LangGraph	7 Hours	Graph state, agents, routing, human-in-loop, multi-agent workflows
MCP Integration	3 Hours	MCP architecture, server/client, tool integration
UiPath + AI Integration	4 Hours	Python scripts, REST API calls, robot orchestration, workflow integration
Enterprise Capstone	3 Hours	End-to-end insurance claims automation solution

5. Detailed 8 Half-Day Training Plan

Day 1: Python for AI Automation - Foundation for UiPath Developers

Concepts / Topics	Hands-on Labs
• Python execution model for automation developers • Variables, functions, modules, classes, and reusable utilities • File handling, CSV, Excel, JSON, and exception handling • REST API calling using Python requests • Insurance data models: policy, customer, claim, premium, and payment records	• Read insurance policy master from Excel • Parse customer and claim JSON files • Create reusable claim validation functions • Call a mock policy verification API • Generate a clean claim-processing output file

Assignment / Outcome: Build an Insurance Document Processor using Python.

Day 2: Generative AI & Prompt Engineering for Insurance Automation

Concepts / Topics	Hands-on Labs
• LLM fundamentals for automation teams • Prompt engineering patterns for enterprise use cases • System, user, and developer-style instructions • Structured output and JSON response design • Insurance use cases: claim summary, policy summary, customer communication, underwriting notes	• Generate claim summaries from raw claim notes • Create a policy coverage explanation • Generate professional customer emails • Extract risk indicators from claim descriptions • Produce JSON output for UiPath consumption

Assignment / Outcome: Create a prompt library for insurance automation scenarios.

Day 3: Python + LLM APIs for AI Automation

Concepts / Topics	Hands-on Labs
• OpenAI SDK and Azure OpenAI API usage • Authentication, API keys, environment variables, and safe configuration • Response parsing and validation • Error handling, retries, and logging • Designing AI services consumable by UiPath	• Build a claim summarization API client • Create a policy Q&A function • Generate classification labels for claims • Parse LLM output into JSON • Create a FastAPI endpoint for AI claim assistance

Assignment / Outcome: Build an Insurance Helpdesk Bot backend using Python and LLM APIs.

Day 4: LangChain for Insurance Knowledge Assistants

Concepts / Topics	Hands-on Labs
• LangChain overview and use cases • Prompt templates and chains • Output parsers and structured responses • Tools and simple agents • Insurance knowledge assistant design	• Build a policy FAQ assistant • Build a customer support assistant • Create a claim status assistant • Use output parsers for structured responses • Create a reusable chain for policy summary generation

Assignment / Outcome: Build an Insurance Policy Assistant using LangChain.

Day 5: Agentic AI with LangGraph - Insurance Claims Workflow

Concepts / Topics	Hands-on Labs
• Agentic AI concepts for automation engineers • LangGraph state, nodes, edges, and conditional routing • Single-agent versus multi-agent design • Agent responsibilities and workflow	• Create a claim intake agent • Create a document validation agent • Create a policy lookup agent • Create a fraud risk review agent • Route claims based on policy and risk status

boundaries • Insurance claim workflow decomposition	

Assignment / Outcome: Build a LangGraph-based Insurance Claim Workflow.

Day 6: Advanced LangGraph - Human Review, Error Recovery and Multi-Agent Flow

Concepts / Topics	Hands-on Labs
• Human-in-the-loop review patterns • Approval, rejection, escalation, and rework routing • Memory and state updates • Retry and error recovery strategies • Multi-agent collaboration for complex insurance workflows	• Build human review checkpoint for high-value claims • Create escalation flow for missing documents • Add retry logic for API failures • Create approval and rejection branches • Build medical / motor / travel claim routing

Assignment / Outcome: Build a Multi-Agent Claim Settlement Workflow with human review.

Day 7: MCP and UiPath Integration for AI Automation

Concepts / Topics	Hands-on Labs
• MCP fundamentals and enterprise relevance • MCP server, MCP client, tools, and tool discovery • Connecting AI agents to business tools and APIs • UiPath integration patterns with Python and REST APIs • Security and governance considerations for AI tool access	• Create a simple MCP tool server concept • Expose policy lookup as an AI tool • Expose claim validation as an AI tool • Invoke Python AI service from UiPath • Trigger UiPath robot from an AI-assisted workflow

Assignment / Outcome: Build an AI Agent that invokes UiPath-supported insurance automation steps.

Day 8: Enterprise Capstone - AI-Powered Insurance Claims Automation

Concepts / Topics	Hands-on Labs
• Capstone architecture design • End-to-end workflow integration • Testing, validation, and presentation preparation • Responsible AI, auditability, and governance review • Capstone presentation and viva	• Integrate Python data processing • Use Generative AI for summarization and classification • Use LangGraph for claim routing and agent orchestration • Use MCP concept for tool access • Connect with UiPath workflow for final automation execution

Assignment / Outcome: Final Capstone: AI-Powered Insurance Claims Automation Platform.

6. Hands-on Projects Covered During Training

- Insurance Policy Reader
- AI Claim Summarizer
- AI Email Generator
- Policy Q&A Assistant
- Insurance Customer Support Agent
- Claim Validation Agent
- Fraud Risk Review Agent
- Medical Claim Review Assistant
- LangGraph Multi-Agent Claim Workflow
- MCP Tool Server Concept
- UiPath + Python + AI Integration
- End-to-End Insurance Claims Automation Capstone

7. Final Capstone Workflow

Step	Workflow Activity
1	Customer uploads claim document
2	Python extracts and validates structured data
3	Generative AI summarizes claim and policy context
4	Policy verification agent checks coverage
5	Fraud review agent identifies risk indicators
6	LangGraph routes claim to approval, rejection, or human review
7	MCP-style tools expose policy lookup, claim validation, and document analysis
8	UiPath robot executes downstream enterprise automation
9	System generates customer email and audit report

8. Lab Setup Requirements

Category	Requirements
Hardware	16 GB RAM minimum, Intel i5 / Apple M1 or above, stable internet
Core Software	Python 3.11+, VS Code, Git, Docker Desktop, UiPath Studio
AI Access	OpenAI API or Azure OpenAI access
Python Libraries	openai, langchain, langgraph, requests, fastapi, pydantic, pandas, openpyxl, faiss-cpu or chromadb
Optional Tools	n8n, Postman, cloud account access for deployment discussion
Sample Data	Policy master, customer master, claims data, medical claim records, motor claim records, customer emails, policy documents

9. Assessment Strategy

Component	Weightage
Daily Hands-on Labs	30%
Assignments	20%
Daily Quizzes / Checks	15%
Participation and Design Discussion	10%
Final Capstone Presentation and Viva	25%

10. Training Deliverables

- Course material and slide references
- Hands-on lab guides
- Python source code examples
- Prompt library for insurance use cases
- LangChain and LangGraph sample workflows
- MCP integration reference example
- UiPath integration guidance
- Capstone project template

- Assessment report and completion certificate

Recommended Positioning: This program should be presented as a practical transition path from traditional RPA development to enterprise AI Automation Engineering for Insurance. The strongest value proposition is the combination of UiPath skills with Python, Generative AI, LangGraph-based agent workflows, MCP-style tool integration, and insurance-specific capstone implementation.